

MUSTAQUEEM FAREEDI

✉ mustaqueemfareedi5@gmail.com ☎ [+91 8095161401](tel:+918095161401)

EDUCATION

Bachelor of Engineering, *New Horizon College of Engineering*

Electrical and Electronics Engineering

CGPA: 6.76

Marthahalli

Bangalore ,India

PUC Education, *Sharanappa meti College*

Percentage: 81%

Lingsugur

Di.Raichur,

Karnataka, India

Secondary Education, *Sharana Basawehwar School (SBR)*

Percentage: 83.36%

Surapur

Di.yadgir

Karnataka, India

SKILLS

Technical

Embedded System,

VLSI and PLC

Programming

Tools

Arduino IDE, Unity Pro 5 and

MATLAB/SIMULINK

Web Technologies

React Js and Node Js

Soft Skills

Teamwork, Time Management, Problem Solving.

Electrical

Renewable Energy, EV Technology and

Power Electronics

PROJECTS

SOLAR-INTEGRATED ADAPTIVE LOAD SHARING FOR MULTI-VEHICLE EV CHARGING STATION USING ESS AND IOT | MATLAB/ SIMULINK, ESP32, IOT-COMMUNICATION |

- Developed a MATLAB/Simulink-based multi-EV charging system integrating Solar PV, Energy Storage System (ESS), and grid supply.
- Designed an adaptive load-sharing strategy to dynamically distribute available charging power among multiple EVs while maintaining the grid power limit.
- Implemented coordinated PV–Grid–ESS power management, enabling ESS charging during excess solar generation and discharging during peak EV demand.
- Developed an IoT-based monitoring system using ESP32, Node.js, and React to visualize EV charging power, ESS SOC, solar generation, and grid parameters in real time.
- Simulated a four-EV charging scenario and evaluated power-flow management under varying solar generation and charging demand.

SELF-CLEANING SOLAR PANEL | ESP32 MICROCONTROLLER & BLYNK IOT|

- Designed and developed an **automated self-cleaning mechanism** for solar panels to reduce dust accumulation and improve power generation efficiency.
- Implemented the system using **ESP32 microcontroller**, integrating sensors and motor drivers for real-time control.
- Developed embedded firmware using **Arduino IDE (Embedded C)** for automation logic and device control
- Integrated **Blynk IoT platform** to enable **remote monitoring, manual override, and real-time status visualization** through a mobile application.
- Utilized **Wi-Fi communication** for seamless data transmission between hardware and cloud platform.
- The project demonstrates hands-on experience in **embedded systems, IoT architecture, and renewable energy applications**.

ALZHEIMER'S DISEASE DETECTION USING MACHINE LEARNING

- Developed a Python-based machine learning model for Alzheimer's disease Prediction
- Performed data preprocessing and built a simple prediction interface.
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- Evaluated the model output to identify patterns associated with different stages of Alzheimer's disease.

GPS TRACKING SYSTEM USING ARDUINO

- Built a real-time GPS tracking system using Arduino and a GPS module..
- Displayed location coordinates for tracking applications.
- Integrated the GPS module with Arduino to process and retrieve real-time location data for tracking applications.

Certificates

NPTEL – Internet of Things

Completed certified course on internet of things concepts and architecture.

Problem Solving and Ideation

Workshop

participated in workshop in Association with artificial intelligence and machine learning